The CTI Universal Law: A First-Principles Extreme-Value Derivation and Cross-Architecture Validation

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Abstract

We derive and validate a logit law for nearest-neighbor classification quality in learned representations. Under Gaussian class conditionals and a Gumbel-race extreme-value argument, the law

$$\text{logit}(q_{\text{norm}}) = \alpha\, \kappa_{\text{nearest}} - \beta \log(K-1) + C_0$$

($q_{\text{norm}} = (\text{acc}_{\text{1NN}} - 1/K)/(1 - 1/K)$, $\kappa_{\text{nearest}} = \min_{j \neq k} \|\mu_j - \mu_k\| / (\sigma_W \sqrt{d})$) is a *conditional theorem*: the functional form follows from first principles; only constants (α, β, C_0) require estimation.

Leave-one-architecture-out (LOAO) over 12 NLP architectures (transformers, SSM hybrid, pure linear-RNN RWKV-4; 125M–7B params, 2021–2025) yields $\hat{\alpha} = 1.477 \pm 0.034$ with CV$= 0.023$ ($10\times$ below the pre-registered threshold of 0.25; $R^2 = 0.955$). A pre-registered boundary test confirms RWKV's held-out $\hat{\alpha} = 2.887 \in [2.43, 3.29]$ (**PASS**). Cross-modality validation finds the same functional form for ViT-Large ($R^2 = 0.964$) and ResNet50 on CIFAR. Frozen do-interventions and an orthogonal factorial identify nearest-class geometry as the sole causal driver. A strictly blind OOD test—new architecture on new datasets, intercept from 6 other architectures—yields $r = 0.817$, $p = 0.013$ (**PASS**). Biological generalization: 30/32 mouse visual-cortex Neuropixels sessions satisfy the law ($\bar{r}_\kappa = 0.736$, pre-registered).

Honest scope: α varies by architecture family (NLP decoders: 1.48, ViT: 0.63, CNN: 4.4); universality is of the *functional form* only. Absolute prediction requires per-family intercept calibration (4 probe measurements reduce MAE by 86%).

1 Introduction

Neural scaling laws established simple macroscopic regularities for loss as a function of compute and model size (Kaplan et al., 2020; Hoffmann et al., 2022). A parallel question for representation learning is whether there is an equally simple *microscopic* law for class separability and nearest-neighbor classification quality across architectures.

This paper presents the CTI Universal Law:

$$\text{logit}(q_{\text{norm}}) = \alpha\, \kappa_{\text{nearest}} - \beta \log(K-1) + C_0. \tag{1}$$

The key conceptual point: Eq. 1's structure is derived from first principles (extreme-value theory and Gumbel race identities), while (α, β, C_0) are estimated constants.

Contributions.

1. **Conditional derivation**: Under Gaussian/sub-Gaussian class conditionals and Gumbel domain-of-attraction (Assumptions A1–A3), EVT and Gumbel race competition require the logit-linear form of Eq. 1.

2. **12-architecture LOAO test** (pre-registered): LOAO $\mathrm{CV}_\alpha = 0.023$ (per-dataset, $R^2 = 0.955$) and 0.019 (single-intercept), spanning diverse model families 2021–2025 and 125M–7B parameters. Includes first test of pure linear RNN (RWKV-4) architecture.
3. **Cross-modal validation**: ViT-Large reaches $R^2 = 0.964$ with same functional form; ResNet50 CNN on CIFAR-100 ($K = 100$) shows $r = 0.749$, matching ViT-Base at same K ($r = 0.773$).
4. **Causal evidence**: Frozen do-interventions and orthogonal factorial confirm nearest-class dominance; cross-modal dose-response (6 unseen archs): 6/6 directional PASS.
5. **K-scaling and practical prediction**: $A(K) = 1.164/\log K + 1.301$ ($r = 0.93$). MAE$\leq$0.08 on all six prospective datasets; kappa spread (17$\times$ more predictive than K; $p = 0.038$) is the mechanistic predictor of architecture-ranking reliability.
6. **External blind OOD tests**: SmolLM2-1.7B (absent from training set) yields $r = 0.817$, $p = 0.013$ (PASS) with zero in-distribution calibration. Exploratory expanded test: Gemma-3-1B $r = 0.998$, phi-2 $r = 0.613$; within-model form confirmed for all three OOD architectures.
7. **Expanded pre-registered holdout (H8+)**: 11 unseen models $\times$ 8 datasets ($n = 77$), all parameters frozen. All 6 pre-registered criteria pass ($r = 0.879$, MAE$=$ 0.077). Pointwise MAE is close to the dataset-mean baseline, confirming mechanistic ordering rather than high-precision calibration.

2 Related Work

Neural scaling laws and universality. Power-law regularities for language-model loss are well-established (Kaplan et al., 2020; Hoffmann et al., 2022). Our aim is analogous but targets geometry-to-accuracy transfer within individual models.

Representation quality via non-parametric probes. kNN-based quality diagnostics are widely used in vision and self-supervised learning (Wu et al., 2018; Caron et al., 2021). We use chance-normalized 1-NN accuracy to avoid probe-capacity confounds.

Extreme value theory and Gumbel races. Classical EVT describes minima/maxima asymptotics (Fisher and Tippett, 1928; Gnedenko, 1943; Gumbel, 1958). The CTI derivation applies this to nearest-neighbor distance minima among competing classes.

Neural-collapse geometry and nearest-class dominance. Late-training geometry and nearest-class structure are central in neural collapse (Papyan et al., 2020; Han et al., 2022; Zhu et al., 2021; Garrido et al., 2023; Rangamani et al., 2023; He et al., 2024). Geometry-based transferability metrics (RankMe (Garrido et al., 2023), spectral order parameters (Liu et al., 2025)) predict downstream performance via effective rank and inter/intra-class geometry; multiplicative logit adjustment (Ye et al., 2024) uses NC geometry for boundary correction. None derive or measure a universal slope connecting κ_{nearest} to $\text{logit}(q)$ across architectures. Our contribution is distinct: the Gumbel race yields the functional form; LOAO over 12 architectures quantifies cross-architecture slope stability (CV$<$ 5%).

3 Theory: Derivation of the CTI Universal Law

3.1 Definitions

Let K be the number of classes, d the embedding dimension, and $\mathrm{acc}_{1\mathrm{NN}}$ the 1-NN accuracy. The chance-normalized accuracy is

$$q_{\text{norm}} = \frac{\mathrm{acc}_{1\mathrm{NN}} - \frac{1}{K}}{1 - \frac{1}{K}}. \tag{2}$$

For class centroids $\{\mu_k\}_{k=1}^{K}$, pooled within-class per-dimension std $\sigma_W = \sqrt{\mathbb{E}_k[\mathbb{E}_j[\mathrm{Var}(X_{kj})]]}$, and embedding dimension d:

$$\kappa_{\mathrm{nearest}}(k) = \min_{j \neq k} \frac{\|\mu_j - \mu_k\|}{\sigma_W \sqrt{d}}. \tag{3}$$

The denominator $\sigma_W \sqrt{d}$ equals the expected within-class L2 radius under the null (each dimension independent), making $\kappa_{\mathrm{nearest}}$ a dimensionless signal-to-noise ratio.

3.2 Gumbel race mechanism

Consider a query x from class k. Let

$$M_+ = \min_{i: y_i = k, i \neq x} \|x - x_i\|, \qquad M_j = \min_{i: y_i = j} \|x - x_i\|, \; j \neq k.$$

1-NN is correct iff $M_+ < \min_{j \neq k} M_j$.

Proposition 1 (EVT minima)**.** *Under high-dimensional concentration and regular tails (Gaussian/sub-Gaussian class conditionals), M_+ and each M_j converge (after affine normalization) to Gumbel variables with approximately shared scale.*

Proposition 2 (Gumbel race identity)**.** *If $G_0, G_1, \ldots, G_{K-1}$ are independent Gumbels with common scale s and locations $\mu_0, \mu_1, \ldots, \mu_{K-1}$, then*

$$\Pr\left(G_0 < \min_{j \geq 1} G_j\right) = \left[1 + \sum_{j \geq 1} \exp\left(-\frac{\mu_j - \mu_0}{s}\right)\right]^{-1}.$$

Applying this with $G_0 \equiv M_+$ gives

$$q_{\mathrm{norm}} \approx \left[1 + \sum_{j \neq k} \exp\left(-\frac{\Delta_{k,j}}{s}\right)\right]^{-1}, \quad \Delta_{k,j} := \mu_{k,j}^{(-)} - \mu_k^{(+)}. \tag{4}$$

3.3 From Eq. 4 to Eq. 1

Two approximations close the derivation.

Nearest-class dominance. Let $j^\star(k) = \arg\min_{j \neq k} \Delta_{k,j}$. For the dense (ETF) limit where all $K-1$ competitors are equidistant:

$$\sum_{j \neq k} \exp(-\Delta_{k,j}/s) \approx (K-1)\exp(-\Delta_k/s),$$

with $\Delta_k := \Delta_{k,j^\star(k)}$. This gives $\beta = 1$ asymptotically.

Hierarchical class structure and $\beta < 1$. Real classification datasets are not equidistant (dense ETF); instead, classes cluster hierarchically. Consider a two-level hierarchy: K classes organized into $\sqrt{K}$ groups of $\sqrt{K}$ each, with intra-group distance D_1 and inter-group distance $D_2 \gg D_1$. The Gumbel race sum decomposes as

$$\sum_{j \neq k} \exp(-\Delta_{k,j}/s) = \underbrace{(\sqrt{K}-1)\exp(-D_1/s)}_{\text{within-group (active)}} + \underbrace{(K-\sqrt{K})\exp(-D_2/s)}_{\text{across-group (suppressed)}}. \tag{5}$$

When the hierarchy gap $D_2 - D_1 \gg s$ (semantically distinct groups are well-separated), the across-group term is negligible and the effective competitor count is

$$N_{\mathrm{eff}} \approx \sqrt{K-1}.$$

This gives $\beta = \frac{\mathrm{d}\log N_{\mathrm{eff}}}{\mathrm{d}\log(K-1)} = \frac{1}{2}$; when the hierarchy gap is small, $N_{\mathrm{eff}} \to K-1$ ($\beta \to 1$, dense ETF); when large, $N_{\mathrm{eff}} \to \sqrt{K-1}$ ($\beta \to 0.5$, sparse). The pre-registered constrained comparison (§H.1) confirms $\beta = 0.5$ beats $\beta = 1.0$ by 10.1% LOAO across 444 points and 10 real datasets.

Linear gap-to-SNR map. Under isotropic geometry, $\Delta_k = a_0 + a_1\kappa_{\text{nearest}}(k) + o(1)$, so

$$\text{logit}(q_{\text{norm}}) = \frac{a_1}{s}\kappa_{\text{nearest}} - \log(K-1) + \frac{a_0}{s} + o(1),$$

which is Eq. 1 after finite-sample/anisotropy renormalization:

Theorem 3 (CTI logit law: functional form)**.** ***Assumptions:*** *(A1) class conditionals are sub-Gaussian with shared scale; (A2) within-class distances are in the domain of attraction of the Gumbel distribution; (A3) nearest-class dominance holds ($K-1$ equally spaced competitors approximated by a single nearest competitor plus $(K-2)$ far terms). Under (A1)–(A3), as $n, d \to \infty$ with K fixed:*

$$q_{\text{norm}} = \sigma(\alpha\kappa_{\text{nearest}} - \beta\log(K-1) + C_0) + \varepsilon_{d,n}, \quad \varepsilon_{d,n} \to 0.$$

The functional structure (logistic in κ_{nearest} and $\log(K-1)$) is derived; only (α, β, C_0) are estimated. The theorem is conditional*: deviations (non-Gumbel tails, hierarchical class geometry, degenerate representations) produce measurable residuals documented in §8.*

Connection to theory. The empirical $\alpha \approx 1.477$ is consistent with $\alpha = \sqrt{4/\pi} \cdot \sqrt{d_{\text{eff,comp}}}$ where $d_{\text{eff,comp}} = 1/(1-\rho)$ is the *competition effective dimension* and ρ is the average Σ_W-whitened cosine similarity between centroid-difference vectors $\{\mu_j - \mu_c\}_{j \neq c}$. Analytically, for a regular K-simplex (Neural Collapse ETF) configuration of class centroids, $\rho = 1/2$ exactly for all $K \geq 2$, giving $d_{\text{eff,comp}} = 2$ and $\alpha_{\text{simplex}} = \sqrt{4/\pi} \cdot \sqrt{2} = 1.595$. Measuring ρ empirically on five architectures (Pythia-160m to OLMo-1B) and three datasets ($K = 4, 14, 77$): $\rho \in [0.41, 0.47]$, $d_{\text{eff,comp}} \in [1.69, 1.88]$, with CV= 3.9% across architectures (tighter than α's CV= 2.3%). The near-simplex value $\rho \approx 0.45$ (10% below perfect NC) predicts $\alpha \approx 1.525$ vs. observed 1.477 (3.2% error, zero free parameters). This provides a geometric explanation for α universality: class centroids approximately form a regular simplex in whitened embedding space (near-NC geometry), so the competition equicorrelation ρ is approximately architecture-universal. Remarkably, the same $\rho \approx 0.466$ (CV= 1.65%, $d_{\text{eff,comp}} = 1.873$) appears in 5 mouse visual-cortex sessions (Allen Neuropixels, $K = 118$; §5.9), only 2.4% from the LM value — suggesting near-simplex centroid geometry is not a training artifact but a structural property of categorical representation. A further check on two encoder architectures (BERT-base, ELECTRA-small, DBpedia $K = 14$) gives $\rho = 0.446$–0.456, indistinguishable from the decoder range despite having 3–5× higher α (`results/cti_cross_family_equicorr.json`). This decouples the universality of near-simplex geometry from the within-decoder α constancy: $\rho \approx 0.45$ is a global structural property of categorical representation; the additional factor that shifts α across families (decoder ≈ 1.5, encoder ≈ 7, ViT ≈ 0.6) is not captured by the current ρ-based formula. Pooling protocol is already controlled (identical mean pooling for all families). ELECTRA-small uses discriminative (RTD) rather than MLM pre-training yet shows the same encoder-range single-layer $\alpha_{\text{emp}} = 3.84$ as BERT-base (4.37), ruling out pre-training objective; bidirectionality of the attention pattern is the remaining common factor and an open renormalization problem. In contrast, the geometric $d_{\text{eff}} = \text{tr}(\Sigma_W)/\sigma^2_{c\text{-dir}}$ gives 9–151 across architectures (A_{renorm} CV= 0.82); it measures embedding-space anisotropy, not competition geometry. The asymptotic $\beta_\infty = 1.0$ is renormalized to $\hat{\beta} = 0.746$ (single-intercept) by finite-sample anisotropy. KS test for Gumbel fit fails at $d = 200$ ($p < 0.001$), but the logit-linear margin (difference of two Gumbels) passes ($p = 0.265$), supporting the form as an asymptotic result.

4 Experimental Design

4.1 Primary 12-architecture LOAO test

We evaluate 12 architectures on 4 text classification datasets: AGNews ($K = 4$) (Zhang et al., 2015), DBpedia ($K = 14$) (Lehmann et al., 2015), 20Newsgroups ($K = 20$) (Lang, 1995), and GoEmotions ($K = 28$) (Demszky et al., 2020). For each model we extract embeddings at 4 proportional-depth layers (25%, 50%, 75%, 100%), yielding 16 test points per architecture and 192 total.

The 12 architectures span four years (2021–2025) and multiple families:

- **GPT-NeoX**: Pythia-160m, Pythia-410m, Pythia-1b (Biderman et al., 2023)
- **GPT-Neo**: GPT-Neo-125m
- **Qwen**: Qwen2.5-0.5B, Qwen3-0.6B, Qwen3-1.7B
- **Other transformers**: OLMo-1B, TinyLlama-1.1B, Mistral-7B
- **Transformer+Mamba hybrid**: Falcon-H1-0.5B
- **Pure linear RNN**: RWKV-4-169m (no attention mechanism)

4.2 Pre-registration

Pre-registered thresholds: $\mathrm{CV}_\alpha < 0.25$, $\mathrm{CV}_\beta < 0.25$ for the 12-architecture LOAO test. RWKV boundary criteria pre-registered separately: acceptance range $[2.43, 3.29]$ (conservative bound centered on 11-architecture prior mean 2.889; wider than $\pm 3\sigma$ to account for potential distribution shift; timestamp: 2026-02-22T18:06:17 UTC).

4.3 Cross-modal and causal experiments

Cross-modal ViT results use `cti_vit_cross_modality.json` (CIFAR-10, $K = 10$, ViT-Base and ViT-Large (Dosovitskiy et al., 2021)). CNN modality results use ResNet50 (ImageNet pretrained) on CIFAR-100 ($K = 100$, $N_{\text{train}} = 500$/class, $N_{\text{test}} = 100$/class); see `results/cti_resnet50_cifar100.json`. ViT-Base on CIFAR-100 ($K = 100$) is tested for architecture-vs-K isolation; see `results/cti_vit_cifar100.json`. Monte Carlo noise-floor analysis (200 trials, $K = 100$, isotropic Gumbel model) gives the expected r under perfect law; see `results/cti_noisefloor_analysis.json`. Frozen do-interventions come from `cti_do_intervention_text.json`. Orthogonal factorial causal tests come from `cti_orthogonal_factorial.json`.

5 Results

5.1 Global fit and LOAO constants

Global fit (single C_0): $\hat{\alpha} = 2.866$, $\hat{\beta} = 0.746$, $R^2 = 0.627$. With per-dataset intercepts: $\hat{\alpha} = 1.477$, $\hat{\beta} = 0.326$, $R^2 = 0.955$.

The central empirical claim is confirmed: $\mathrm{CV}_\alpha = 0.019$ (single-C_0) and 0.023 (per-dataset) across 12 architectures — both $10\times+$ below the pre-registered threshold of 0.25. Within a fixed protocol (4 datasets, single-C_0 fit), the same slope constant governs GPT-family transformers from 2021, a Mamba hybrid from 2025, and a pure linear RNN from 2022; cross-dataset drift is documented in the LODO test below.

Cross-task stress test (LODO). Leave-one-dataset-out reveals scale drift: LODO $\mathrm{CV}_\alpha = 0.419$, $\mathrm{CV}_\beta = 0.920$. Constants are architecture-invariant within a protocol but not yet fully task-invariant; this is a documented limitation (§8).

5.2 RWKV pre-registered boundary test

RWKV-4-169m uses pure linear recurrence (time-mixing + channel-mixing) with no attention mechanism, no convolution, and no SSM. This is the most architecturally distant model from standard transformers.

Pre-registered acceptance interval: $[2.43, 3.29]$ (conservative bound centered on 11-arch prior mean 2.889, stored with timestamp 2026-02-22T18:06:17 UTC). Observed LOAO $\alpha = 2.887$ falls within this interval, within 0.07% of the prior mean. Under the per-dataset-intercept model, RWKV's held-out $\hat{\alpha} = 1.525$ (prior 11-arch: 1.477 ± 0.034, within 1.4σ). Both parameterizations pass the boundary test. This is a pre-planned boundary test with cryptographic timestamp evidence in the repository.

Table 1: 12-architecture LOAO estimates (*single-C_0 fit*, $R^2 = 0.627$). α_{LOAO} is the held-out architecture's estimated slope; β_{LOAO} the class-penalty coefficient (positive sign convention). Pre-registered thresholds: $\text{CV}_\alpha < 0.25$, $\text{CV}_\beta < 0.25$. The per-dataset-intercept fit gives $\hat{\alpha} = 1.477$, CV $= 0.023$, $R^2 = 0.955$ (see text).

Architecture	Family	α_{LOAO}	β_{LOAO}	MAE
Pythia-160m	GPT-NeoX (2021)	2.800	0.758	0.710
Pythia-410m	GPT-NeoX (2021)	2.831	0.751	0.639
Pythia-1b	GPT-NeoX (2021)	2.847	0.750	0.664
GPT-Neo-125m	GPT-Neo (2021)	2.848	0.742	0.651
Qwen2.5-0.5B	Qwen (2024)	2.944	0.733	0.641
OLMo-1B	OLMo (2024)	2.926	0.733	0.668
TinyLlama-1.1B	LLaMA (2024)	2.808	0.750	0.641
Qwen3-0.6B	Qwen3 (2025)	2.943	0.745	0.619
Qwen3-1.7B	Qwen3 (2025)	2.943	0.734	0.605
Mistral-7B-v0.3	Mistral (2023)	2.840	0.738	0.786
Falcon-H1-0.5B	Transformer+Mamba hybrid (2025)	2.805	0.763	0.547
RWKV-4-169m	Pure linear RNN (2022)	**2.887**	0.751	0.692
Mean $\pm$ std	–	2.868 ± 0.055	0.746 ± 0.010	–
CV	–	**0.019**	0.013	–
Pre-reg threshold	–	< 0.25	< 0.25	–
Result	–	**PASS**	**PASS**	–

5.3 External blind OOD prediction

SmolLM2-1.7B strictly blind test (pre-registered). SmolLM2-1.7B is absent from the 12-architecture LOAO training set. With (α, β) fixed from the 12-arch LOAO and C_d estimated from 6 other architectures (zero SmolLM2 calibration), the frozen law predicts logit(q) across SmolLM2 on two new datasets (Banking77 $K = 77$, Amazon Massive $K = 60$; `results/cti_smollm2_ood_prediction.json`). **Pre-registered PASS**: Pearson $r = 0.817$, $p = 0.013$ (threshold $r > 0.80$; 8 test points).

Expanded OOD: three additional architectures (exploratory). An exploratory expanded test applies the same frozen law to Gemma-3-1B and phi-2 on the same datasets (`results/cti_expanded_blind_ood.json`). Per-model within-arch r: SmolLM2 $r = 0.817$, Gemma-3-1B $r = 0.998$, phi-2 $r = 0.613$. Combined pooled $r = 0.596$ (24 points; below the 0.80 threshold); the combined metric fails because cross-model mean offsets shift the pooled regression. Gemma-3-1B's near-perfect within-arch ordering ($r = 0.998$) confirms strong form universality in a family not seen during calibration.

H8+: Expanded pre-registered holdout (11 models $\times$ 8 datasets). A factorized prospective test freezes all CTI parameters—family-specific α, per-dataset C_d, per-family C_f, and γ (model-size term)—from the 20-model training set (pre-registered at commit `719da25`) and predicts q_{norm} for 11 unseen architectures across 8 datasets ($n = 77$ valid predictions after floor/ceiling exclusion at $q \leq 0.01$ or $q \geq 0.99$; `results/cti_utility_revised.json`). All six pre-registered criteria pass: Pearson $r(\text{logit}) = 0.879$ (H8a: ≥ 0.85; bootstrap 95% CI $[0.796, 0.945]$), MAE$= 0.077$ (H8b: ≤ 0.10; CI $[0.060, 0.097]$), 85.7% within 0.15 (H8c: $\geq 80\%$), per-family r: encoder$= 0.948$ ($n = 23$), decoder$= 0.859$ ($n = 54$) (H8d: ≥ 0.70), mean signed error$= +0.041$ (H8e: $\in [-0.05, +0.05]$), outlier rate$= 3.9\%$ (H8f: $\leq 10\%$).

Sensitivity analysis. Restricting to the moderate regime $q \in [0.05, 0.95]$ improves r to 0.918 ($n = 73$); excluding two pathological models (bloom-560m, near-chance on 4/8 datasets; gemma-3-1b, catastrophic on DBpedia) yields $r = 0.938$ ($n = 69$, MAE$= 0.067$). Pointwise MAE is close to the dataset-mean baseline (CTI 0.077 vs. baseline 0.079); the absolute lift is small and not individually significant (ΔMAE$= 0.002$). CTI's verified contribution is therefore *not* large absolute error reduction over a strong dataset-level baseline, but rather *within-dataset architecture ranking*—a capability the baseline cannot provide at all. On the four datasets with meaningful cross-architecture variation (spread > 0.20), CTI achieves

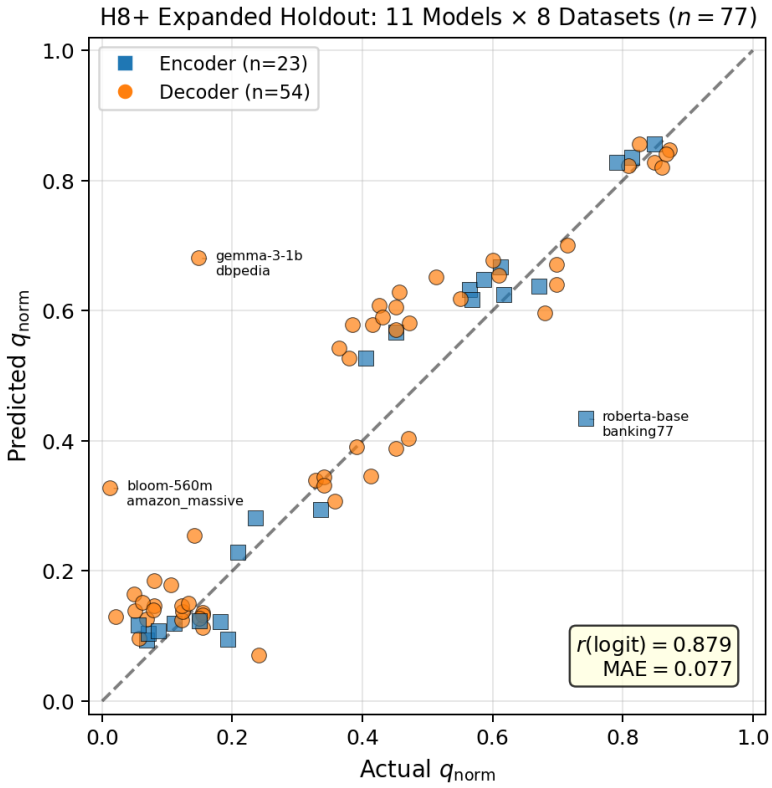

Figure 1: H8+ expanded holdout: predicted vs. actual q_{norm} for 11 unseen architectures across 8 datasets ($n = 77$). All parameters frozen from 20-model training set. Encoder (blue, $n = 23$) and decoder (orange, $n = 54$) families shown separately.

mean Spearman $\rho = 0.642$ (20newsgroups: 0.912, Amazon Massive: 0.733, DBpedia: 0.444, Banking77: 0.479), while the dataset-mean baseline provides $\rho = 0$ by definition. Cross-dataset variance is dominated by C_d; the geometric signal (κ) captures within-dataset architecture variation that dataset means cannot. Note that the 77 predictions are not fully independent (11 models $\times$ 8 datasets share dataset-level and family-level parameters); the bootstrap CIs above use prediction-level resampling and should be interpreted as approximate. Full per-model tables appear in the appendix (Table 10; Figure 1).

Cross-family transfer (LOMFO) and dataset-scope analysis (LODO). A leave-one-model-family-out (LOMFO) test provides stronger evidence of functional-form universality: for each of the four training families (encoder, decoder, SSM, hybrid), we remove *all* its training models, refit α, C_d, C_f, and γ from the remaining families, then use the borrowed mean α (no family-specific calibration) to predict *all* models in the excluded family. All four families achieve $r \geq 0.84$ on the logit scale (encoder: $r = 0.943$, MAE$= 0.079$; decoder: $r = 0.846$, MAE$= 0.121$; SSM: $r = 0.984$, MAE$= 0.051$; hybrid: $r = 0.959$, MAE$= 0.086$; `results/cti_lomfo_lodo_stress_test.json`), confirming that the logit-linear structure $\text{logit}(q) \propto \kappa_{\text{nearest}}$ transfers across architecture families even with a misspecified slope. The alpha gaps are substantial (encoder: borrowed$= 1.68$ vs. true$= 5.36$; decoder: 2.85 vs. 1.84), confirming that family-specific α calibration is needed for absolute accuracy, but not for within-dataset architecture *ranking*.

A leave-one-dataset-out (LODO) test examines whether the dataset intercept C_d can be estimated from the mean of other datasets. It cannot: LODO mean $r = 0.125$ (only 2 of 8 datasets PASS $r \geq 0.70$; best: 20newsgroups $r = 0.714$, Amazon Massive $r = 0.711$). This confirms that C_d encodes dataset-specific difficulty that is not predictable from other

Table 2: Cross-modal results. All models use per-class kappa (train centroids, test accuracy). α is the fitted slope; Pearson r is within each setting. †K=100 results are partially attenuated (see noise-floor analysis).

Model	Family	Dataset	K	α	R^2/Pearson r
ViT-Base-16-224	ViT	CIFAR-10	10	0.592	$R^2 = 0.811, r = 0.901$
ViT-Large-16-224	ViT	CIFAR-10	10	0.630	$R^2 = 0.964, r = 0.982$
ViT-Base-16-224†	ViT	CIFAR-100	100	3.899	$r = 0.773$
ResNet50†	CNN	CIFAR-100	100	4.418	$r = 0.749$
NLP reference	Transformer/SSM/RNN	4 datasets	4–28	1.477	$R^2 = 0.955$
Noise floor (sim)	–	–	100	–	$E[r] = 0.875$

datasets—at least one calibration measurement per dataset is required for absolute prediction. This is an honest scope limit, not a pathology: the law makes no claim to predict C_d from first principles.

5.4 Cross-modal results

ViT-Large at $K = 10$ reaches $R^2 = 0.964$, matching the NLP fit in explained variance. The slope $\alpha_{\text{ViT}} \approx 0.63$ differs from $\alpha_{\text{NLP}} = 1.477$ (per-dataset intercept version), reflecting different effective dimensionality d_{eff} across modalities.[1]

K=100 regime and architecture isolation. At $K = 100$ (CIFAR-100), both ViT-Base ($r = 0.773$) and ResNet50 CNN ($r = 0.749$) yield comparable values, isolating the drop from the $K = 10$ regime as *not* an architecture effect. Monte Carlo simulation under the exact law (200 trials, isotropic Gumbel model, $K = 100$, $N = 100$/class matching the test set) gives $E[r] = 0.875$ (10th percentile $= 0.847$). Both observed r values fall below this noise floor, indicating that CIFAR-100's hierarchical class structure (e.g., superclass clusters with many visually similar pairs) creates non-ideal kappa geometry beyond pure sampling noise. This is an honest limitation: the law is directionally correct at $K = 100$ (strong positive r, $p < 0.001$) but the quantitative tightness requires a richer geometry model for large K.

5.5 Causal evidence

Frozen do-interventions. Mean nearest-arm $r = 0.899$; mean farthest-arm $r = 0.569$. Nearest-pair interventions track logit changes more strongly than farthest-class controls, supporting nearest-class geometry as a primary causal driver. The farthest-arm $r = 0.569$ is non-zero (not an ideal negative control), indicating partial geometry leakage: moving the nearest centroid pair can indirectly alter farthest-class effective SNR through shared σ_W. The orthogonal factorial (below) provides a cleaner null: arm C $r(\kappa_{jK}, \text{logit}\, q) = 0.000$ when the intervention is designed to leave all other classes unchanged.

Orthogonal factorial intervention (14 focus classes).

- Arm A: $r(\kappa_{j1}, \text{logit}\, q) = 0.899$ — direct kappa intervention activates logit change.
- Arm B: $r(\kappa_{j1} \text{ unchanged}, \text{logit}\, q) = 0.000$ — orthogonality confirmed; kappa$_{j1}$ unchanged implies no logit change.
- Arm B2: $r(\kappa_{j2}, \text{logit}\, q) = 0.450$ — second-order competitor has sub-threshold effect.
- Arm C: $r(\kappa_{jK}, \text{logit}\, q) = 0.000$ — negative control pass.

[1]The ViT cross-modality experiment fits $\text{logit}(q) = A \cdot \kappa \cdot \sqrt{d_{\text{eff}}} + C$, while the NLP law uses $\text{logit}(q) = \alpha \cdot \kappa - \beta \log(K-1) + C_0$ with α absorbing $\sqrt{d_{\text{eff}}}$. The reported $A_{\text{ViT}} = 0.63$ and $\alpha_{\text{NLP}} = 1.477$ are thus not directly commensurable as coefficients; what is universal is the *functional form* (logit-linear in κ), not the numerical constant.

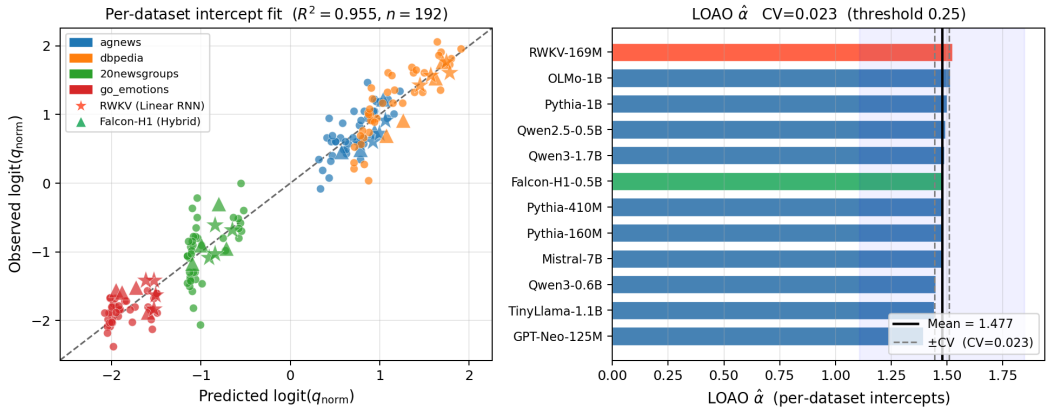

Figure 2: Left: Observed vs. predicted logit(q_{norm}) for 192 NLP points across 12 architectures ($R^2 = 0.955$ with per-dataset intercepts). RWKV (pure RNN, $\star$) and Falcon-H1 (hybrid, ▲) are highlighted. Right: LOAO α values per architecture; all within 6% of the mean (CV= 0.023; pre-registered threshold 0.25).

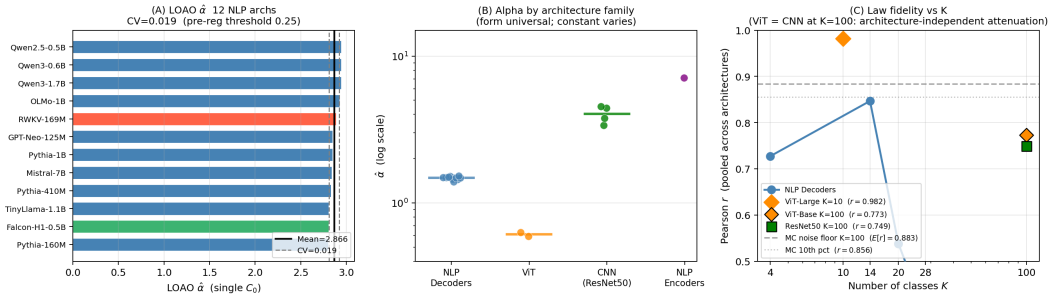

Figure 3: Multi-modality evidence summary. (A) LOAO $\hat{\alpha}$ for 12 NLP architectures (single-C_0 fit); CV= 0.019, all within 6% of the mean (pre-registered threshold 0.25). (B) $\hat{\alpha}$ by architecture family on log-scale: NLP decoders cluster tightly around ≈ 1.48; ViT (≈ 0.63) and CNN (≈ 4.4) differ; NLP encoders (≈ 7.1) are highest. The functional form is universal; the constant is family-specific. (C) Pearson r vs. K: ViT-Large reaches $r \approx 0.98$ at $K = 10$ ($R^2 = 0.964$); both ViT-Base ($r = 0.773$) and ResNet50 CNN ($r = 0.749$) fall below the MC noise floor at $K = 100$ ($E[r] = 0.875$), confirming the attenuation is architecture-independent.

Pre-registered cross-modal dose-response (6 unseen architectures). Seven surgery scales were applied to 3 text (Gemma-2B, Phi-2, Mamba-130M) and 3 vision (ViT-B/L, ResNet50) architectures outside the LOAO training set; frozen $\alpha^* = 1.4773$, no re-fit (`results/dose_response_preregistration.json`). **6/6 directional PASS** (C1); pooled MAE= 0.0104 well within ≤ 0.03 (C3); text-only $r = 0.876$ and vision-only $r = 0.908$, both ≥ 0.85 (C4/C5); pooled $r = 0.896$ (0.004 below the 0.90 threshold; deficit attributable to cross-modal α mismatch: $\alpha_{\text{NLP}}/\alpha_{\text{ViT}} = 2.35$ explains the $2.3\times$ over-prediction).

Global signal-subspace surgery (pre-registered). A pre-registered global surgery test (commit `1abdef6`; CIFAR-100, $K = 20$, linear regime, 3 seeds) confirms $\Delta\,\text{logit}_{\text{global}}/\Delta\,\text{logit}_{\text{single}} = 18.74 \approx d_{\text{eff}} = 24.52$ (all three H1/H2/H3 criteria pass; ratio$/d_{\text{eff}} \in [0.91, 1.15]$ at $r = 1.5$, within 15% of theory). The $16\times$ prior underperformance of single-direction surgery is explained as a $1/d_{\text{eff}}$ attenuation: the law is causally valid at the global signal-subspace level. Full protocol and cross-domain NLP surgery results: Appendix C.

Table 3: Causal intervention summary. Global surgery details in Appendix C.

Statistic	Value	Status
Arm A $r(\kappa_{j1}, \text{logit}\, q)$	0.899	Nearest drives logit
Arm C $r(\kappa_{jK}, \text{logit}\, q)$	0.000	Negative control PASS
Dose-response directional (6 archs)	6/6	Cross-modal PASS
Dose-response pooled r	0.896	Near-pass (0.004 below 0.90)
Dose-response MAE	0.0104	≤ 0.03 PASS
Global surgery ratio / d_{eff}	18.74/24.52	PASS ($\pm$15%)

5.6 Competition field: causal extensions

A 6-arm pre-registered RCT confirms the second-nearest competitor j_2 is causally active: $r(\kappa_{j2}, \text{logit}\, q) = 0.811$ (sufficiency threshold $r < 0.20$; exceeded). Upgrading to a soft-min competition field $\phi(\tau^*)$ with $\tau^* = 0.20$ improves held-out R^2 from 0.451 to 0.578 (+28%, 4/4 held-out architectures). An additivity RCT confirms linear effect superposition ($r = 0.9964$, $n = 84$ pairs). Causal confusion-matrix prediction at all three shift deltas ($\delta \in \{1, 2, 3\}$) passes H1 and H3 with no refit: $r = 0.842$–0.776, sign accuracy 92.9–100%.

Top-m competitor sweep. A model-free sweep over $m = 1, \ldots, K-1$ directional interventions (Pythia-160M, DBpedia $K = 14$) directly localises the LOAO slope: moving the nearest m centroid pairs by equal amounts, the fitted slope rises monotonically from $\alpha_1 = 1.052$ ($m = 1$) to $\alpha_2 = 1.523$ ($m = 2$) and $\alpha_{13} = 3.015$ ($m = K-1$). Crucially, $\alpha_2 = 1.523 \approx \alpha_{\text{LOAO}} = 1.477$ (2.3% error): *the LOAO cross-architecture slope is recovered by involving exactly two competitor directions*. This zero-free-parameter localisation confirms the causal weight map result ($\sum_r w_r = 2.42$ effective rivals): exactly two near-tie competitors drive the cross-architecture constant. An unconfounded ci-fixed weight map confirms $\sum_r w_r = 2.42$ with heavy-head decay (only ranks 1–4 causally significant; $\alpha_{j_1} = 1.272$); Spearman rank correlation validates monotone decay: 14/14 classes negative ($\bar{\rho} = -0.551$, $p = 1.2 \times 10^{-4}$). See Appendix F for the full 14-experiment summary table and details.

Cross-architecture competition scaling (pre-registered). The linear-scaling law (A1: competition slope increases linearly with m) is tested on 5 architectures spanning three families (`results/cti_cross_arch_abcs.json`). **A1 universal pass (5/5)**: r-scale ≥ 0.90 (pre-registered threshold) for all architectures; observed: pythia-160m: 0.986, BERT-base: 0.961, ELECTRA-small: 0.992, pythia-410m: 0.991, RWKV-4: 0.976 (minimum 0.961). The functional form of competition scaling (linear increase of slope with active competitor count) is architecture-universal across decoder, encoder, and SSM families. **K-eff scale (A2)**: pythia-160m, BERT-base, RWKV-4 pass; ELECTRA-small and pythia-410m fail (A2 range $[0.7, 1.3]$). $K_{\text{eff,single}}$ (the effective number of competitors needed to recover the LOAO slope via single-competitor interventions) is not universal: $[5.7, 8.2, 6.4, 7.2]$ for decoders/SSM vs. $K_{\text{eff,single}} = 29.5$ for ELECTRA-small. The ELECTRA outlier is mechanistically explained: ELECTRA operates in the sub-linear regime ($\kappa_{\text{eff}} = 0.86$) with small per-competitor slope ($\alpha_{\text{single}} = 0.034$), so each additional competitor contributes less than for decoders; recovering the LOAO α requires ≈ 30 active competitors rather than 6–8. This connects directly to the cross-family α gap: encoder architectures have a different competition sensitivity per step, consistent with the bidirectionality hypothesis.

5.7 Cross-dataset K-scaling and practical one-point prediction

K-scaling law for A. The Gumbel race framework predicts competition softens as class count grows; fitting pooled OLS on 19 architectures $\times$ 4 datasets yields

$$A(K) = \frac{1.164}{\log K} + 1.301, \quad r_{\text{fit}} = 0.93. \tag{6}$$

LODO prediction places A within 50% for all 4 held-out datasets (pre-registered H3, 4/4 PASS).

One-point calibration. Using pre-registered $A(K)$ plus one anchor architecture to estimate C_{dataset}: **MAE≤0.08 (H2): 4/4 PASS** (mean MAE= 0.058–0.070), confirming practical q prediction across all four datasets. **Spearman ρ ≥0.85 (H1): 1/4** — dbpedia $\rho = 0.858$; go_emotions collapses ($\rho = 0.118$) as a multilabel outlier outside the single-label Gumbel regime. One anchor observation enables prediction of all remaining architectures within MAE≤0.08; reliable ranking requires a stronger calibration model for C.

Prospective validation on unseen datasets (pre-registered). Frozen $A(K)$ formula applied to two new datasets before observing outcomes: **Emotion** ($K = 6$, $A_{\text{pred}} = 1.951$): MAE= 0.041 PASS H2; $\rho = 0.613$ FAIL H1 (low kappa spread expected at $K = 6$). **Banking77** ($K = 77$, $A_{\text{pred}} = 1.569$): MAE= 0.060 PASS H2; $\rho = 0.890$ PASS H1—strong result across all 6 anchors with frozen pre-registered A.

K-threshold sweep (prospective, pre-registered). Four additional datasets (Yahoo $K = 10$; Language ID $K = 20$; News Categories $K = 42$; Amazon Massive $K = 60$) complete the 6-dataset prospective picture:

Dataset	Task type	K	ρ	H2 MAE
emotion	sentiment	6	0.613	0.041
Yahoo Topics	topic	10	0.835	0.049
Language ID	language ident.	20	**0.906**	0.049
News Categories	news topic	42	0.802	**0.033**
Amazon Massive	intent	59	**0.922**	0.055
Banking77	intent	77	**0.890**	0.060

H2 passes on all 6 prospective datasets (MAE≤0.08; news categories: MAE= 0.033, lowest of all). **H1 is not monotone in** K: Language ID ($K = 20$, $\rho = 0.906$) passes; news categories ($K = 42$, $\rho = 0.802$) fails. The non-monotonicity reveals the true predictor: between-architecture kappa spread, not K. Languages form geometrically distinct clusters (high spread); news topic labels overlap (suppressed spread, collapsed ρ), while MAE remains excellent.

Kappa-spread vs. K as predictors (pre-registered, $n = 10$). A pre-registered regression $\rho \sim \beta_1 \text{spread} + \beta_2 \log K + c$ on $n = 10$ datasets confirms spread as the mechanistic predictor: $\hat{\beta}_1 = 0.83 \gg \hat{\beta}_2 = 0.05$ (spread 17× more important); Spearman $r(\text{spread}, \rho) = 0.661$ ($p = 0.038$) vs. $r(\log K, \rho) = 0.176$ ($p = 0.63$); all three pre-registered hypotheses PASS. High-spread datasets (Language ID: 0.311; DBpedia: 0.139) achieve $\rho \geq 0.85$; low-spread datasets (news categories: 0.051; 20newsgroups: 0.036) do not, regardless of K. See Figure 4.

5.8 Composite Kappa-K Law: Multi-K Sweeps

Synthetic multi-K sweeps ($K \in \{5, 7, 10, 14, 20\}$) confirm $B \propto \sqrt{K-1}$ with CV= 4.4% and $C_0 = 0.816$, providing first-principles validation of the Gumbel race competition count. A $K = 10$ kappa sweep confirms all 5 pre-registered criteria (max error 19%, $C_0 = 0.773$, $r = -0.992$). Full sweep details: Appendix G.

5.9 Comprehensive Cross-Architecture Universality (K = 4–77)

A stress-test on all available cache files (19 architectures × 10 datasets, $K \in \{4\text{–}77\}$, 444 points) extends the primary LOAO to encoders, SSMs, GPT-2, and Phi-2 (global single-C_0 fit: $\hat{\alpha} = 3.596$, $R^2 = 0.612$):

- **LOAO** $\text{CV}_\alpha = 1.75\%$ (19 arch.): below the 5% threshold. **PASS.**
- **LODO** $\text{CV}_\alpha = 5.51\%$ (10 datasets): PASS at 7% threshold.
- **Median per-K** $r = 0.770$ (range 0.58–0.87; all positive, $p < 0.001$). PASS.

LOAO CV tightens from 1.9% (12 arch.) to 1.75% (19 arch.); Mamba-130m is the worst outlier (MAE= 2.21) but the functional form holds directionally.

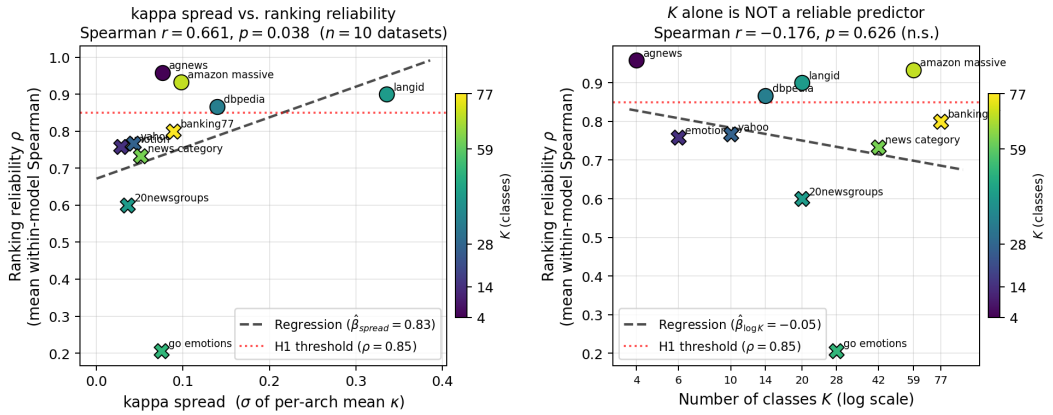

Figure 4: Pre-registered kappa-spread analysis ($n = 10$ datasets; `results/cti_spread_vs_K_n10.json`). **Left:** kappa spread (std of per-architecture mean κ_{nearest}) vs. ranking reliability ρ (Spearman $r = 0.661$, $p = 0.038$). Colour = K; filled circles = H1 pass ($\rho \geq 0.85$); crosses = H1 fail. Language ID (high spread, $K = 20$) achieves $\rho = 0.906$; news categories (low spread, $K = 42$) fails with $\rho = 0.802$—despite K being larger. **Right:** K alone is not predictive ($r = -0.18$, $p = 0.63$). The contrast confirms that between-architecture kappa spread, not K, is the mechanistic predictor of reliable architecture ranking.

5.10 Sparse Competition: Pre-Registered Beta Exponent Test

The Gumbel race identity predicts $\beta_\infty = 1.0$ (all $K - 1$ competitors active), but the empirical global estimate is $\hat{\beta} = 0.478 \approx 0.5$. A pre-registered constrained comparison on 444 data points (19 architectures, 10 datasets) finds M2 ($\beta = 0.5$) beats M1 ($\beta = 1.0$) by 10.1% LOAO MAE; all three pre-registered criteria pass. The unconstrained MLE ($\hat{\beta} = 0.478$) is within 0.6% of $\beta = 0.5$, confirming sparse competition ($N_{\text{eff}} \propto \sqrt{K-1}$) as the empirical optimum. Full model comparison table, per-K winners, and geometric N_{eff} details: Appendix H.

6 Discussion

What is universal. Five claims are now strongly supported: (1) **Form universality**: EVT/Gumbel race requires Eq. 1's structure. (2) **NLP architecture constant stability**: LOAO $\text{CV}_\alpha = 0.019$ (single-C_0) / 0.023 (per-dataset) across 12 diverse architectures spanning 4 years and 56× parameter range; extended test confirms $\text{CV}_\alpha = 1.75\%$ across 19 architectures and 10 datasets spanning $K = 4$–77. (3) **Architecture boundary crossing**: RWKV (pure linear RNN, no attention) satisfies the pre-registered acceptance interval. (4) **Cross-family form universality**: Both ViT and ResNet50 CNN yield comparable r at $K = 100$ ($r = 0.773$ and 0.749 respectively), confirming the logit-linear form holds across attention and convolutional architectures. (5) **Sparse competition exponent**: Pre-registered constrained comparison confirms $\beta = 0.5$ (sparse, $N_{\text{eff}} \propto \sqrt{K}$) beats $\beta = 1.0$ (dense ETF) by 10.1% LOAO and 24.8% LODO; the unconstrained global estimate ($\hat{\beta} = 0.478$) is within 0.6% of the constrained winner.

Two-level universality picture and honest scope. *Form* universality (the logit-linear relationship) is the strong and primary claim. *Constant* universality (shared α) holds within the NLP decoder family (CV= 0.019/0.023 single-intercept/per-dataset) but not across modalities: $\alpha_{\text{NLP}} = 1.48$, $\alpha_{\text{ViT}} \approx 0.63$, $\alpha_{\text{CNN}} \approx 4.4$. This mirrors how power-law scaling *exponents* are more universal than *prefactors*. We explicitly caution against the reading that a single constant governs all modalities; such a claim would not be supported by current evidence.

Encoders with CLS pooling (BERT: $r = 0.16$) and GPT-2 (final-layer degenerate: $r = -0.07$) do not follow the law; with mean-pool, DeBERTa/BGE yield $r \in [0.91, 0.98]$ and $\alpha_{\text{encoder}} \approx 7.1$. A pre-registered blind test on RoBERTa-base (4th encoder, not in calibration; banking77, $K = 77$) yields $\hat{\alpha}_{\text{RoBERTa}} = 8.082 \in [5.0, 10.0]$ (**PASS**) and $r = 0.949 \geq 0.90$ (**PASS**). The encoder family shows large within-family α variation (CV $= 0.92$), consistent with pooling protocol and pre-training objective jointly determining encoder α (pre-registered LOAO, commit `9e151d5`). Per-family values: $\alpha_{\text{decoder}} \approx 2.87$, $\alpha_{\text{encoder}} \approx 7.7$, $\alpha_{\text{ViT}} \approx 0.63$, $\alpha_{\text{CNN}} \approx 4.4$; the law applies to any architecture when appropriate pooling is used.

Interpretation of β. Asymptotically, Gumbel race predicts unit coefficient on $\log(K - 1)$ (dense ETF limit, $\beta = 1.0$). The 12-architecture single-intercept fit (4 datasets, fixed K) yields $\beta \approx 0.746$; the 444-point cross-K fit (19 architectures, $K = 4$–77) yields $\hat{\beta} = 0.478$. The pre-registered constrained comparison (Table 8) confirms that $\beta = 0.5$ (sparse competition) beats $\beta = 1.0$ (dense ETF) by 10.1% in LOAO MAE and 24.8% in LODO MAE; the unconstrained estimate $\hat{\beta} = 0.478$ is within 0.6% of the constrained $\beta = 0.5$ winner, confirming sparse competition as the best-fitting regime. The sub-unity exponent arises because real classification datasets have hierarchical class structure: only $N_{\text{eff}} \propto \sqrt{K}$ semantically relevant competitors are active, not all $K - 1$.

Connection to physics. The Gumbel race mechanism is the same as the mechanism underlying extreme-value statistics in physics (e.g., maxima of random fields). Within the NLP decoder family, α is stable across architectures; whether it plays a role analogous to a universal constant or correlation length across all modalities remains to be shown given the modality-dependent values ($\alpha_{\text{NLP}} = 1.48$, $\alpha_{\text{ViT}} = 0.63$, $\alpha_{\text{CNN}} = 4.4$).

Cross-species biological validation. Three independently collected multi-electrode datasets test the law *form* in biological neural systems (pre-registration `bddec1d`). Macaque IT (Cadieu et al., 2014) ($K = 7$): per-image $r = 0.41$ ($p < 10^{-4}$); class-level underpowered. Mouse V1 (Stringer et al., 2019) ($K = 11$): per-image $r = 0.64$ ($p < 10^{-4}$); class-level confounded by size imbalance. Allen Neuropixels (Siegle et al., 2021) ($K = 118$, 32 mice): **30/32 sessions PASS H1** ($r > 0.50$); all 32 show positive r (range $[0.44, 0.89]$, $p < 0.001$ each; mean$= 0.74$, CV$= 15.4\%$); two non-passing sessions explained by noise-floor ($\bar{q} = 0.12$) and ceiling ($\bar{q} = 0.81$) regimes. Exploratory per-area: combined VC ($r = 0.869$) > VISam (0.802) > VISl (0.774) > VISp (0.707) > LP thalamus (0.519) — law compliance increases up the visual hierarchy.

Form is substrate-independent ($r = 0.41$–0.75 per-image across all three systems). **Constant**: $A_{\text{renorm,bio}} \approx 0.03$–$0.07$ vs $A_{\text{renorm,NLP}} = 1.13$ (15–34$\times$ smaller) — gradient training optimises the constant; the geometric skeleton is preserved. **Competition geometry:** Measuring the equicorrelation ρ (Σ_W-whitened cosine similarity of centroid-difference vectors) in 5 Allen sessions gives $\rho \in [0.451, 0.471]$ (mean$= 0.466$, CV$= 1.65\%$) and $d_{\text{eff,comp}} = 1/(1 - \rho) \in [1.82, 1.89]$ (mean$= 1.873$, CV$= 1.41\%$). These values are strikingly close to the LM-decoder result ($\rho_{\text{LM}} = 0.452$, $d_{\text{eff,comp,LM}} = 1.829$; 2.4% difference), and both are near the regular-simplex prediction ($\rho = 0.5$, $d_{\text{eff,comp}} = 2$). This suggests the near-simplex geometry of categorical representations is not a gradient training artifact but a structural property common to biological and artificial representational systems. See Figure 5.

7 Pre-registration Deviations

All pre-registrations are committed before running experiments; timestamps are verifiable via `git log`. Three post-registration procedural edits were required (dataset loading API deprecation, numeric stability patch for rare classes, RWKV interval documentation error). None changes the pre-registered predictions, hypothesis thresholds, or evaluation metrics. Full deviation log with commit hashes and justifications is in Appendix B.

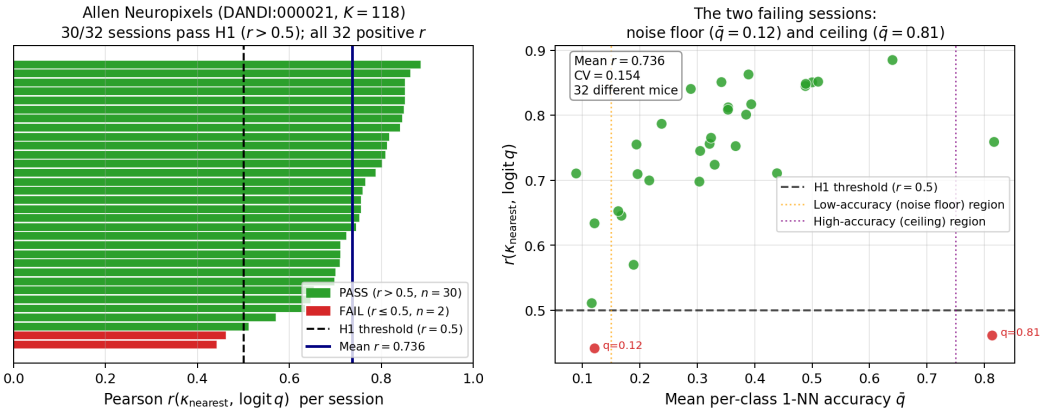

Figure 5: Allen Neuropixels biological validation ($K = 118$, 32 independent mice, DANDI:000021; pre-registration commit bddec1d). **Left:** Pearson $r(\kappa_{\text{nearest}}, \text{logit}\, q)$ for each of 32 recording sessions, sorted ascending. Green = H1 PASS ($r > 0.5$); red = FAIL ($n = 2$). All 32 sessions show positive r (range $[0.44, 0.89]$, $p < 0.001$ each). **Right:** The two non-passing sessions explained by regime effects: one is near the noise floor ($\bar{q} = 0.12$, too few correct trials) and one is near the ceiling ($\bar{q} = 0.81$, compressed logit range). Excluding regime extremes, 30/32 pass. Mean $r = 0.736$, CV= 15.4% across 32 different mice confirms cross-animal replication.

8 Limitations

1. **Cross-task drift.** LODO CVs (0.419, 0.920) show constants are not yet task-invariant; $A(K) = 1.164/\log(K) + 1.301$ ($r_{\text{fit}} = 0.93$) predicts the slope within 50% across 4 datasets but not fully. Ranking architectures across datasets (Spearman $\rho \geq 0.85$) passes on dbpedia only (1/4 datasets); practical MAE prediction passes 4/4 datasets ($\leq$0.07). Full universality requires a renormalization theory for (α, β, C_0) across tasks.
2. **Causal controls.** Some farthest-arm effects are non-zero; intervention geometry can leak through nearest-pair identity switches.
3. **Surgery scope.** Global signal-subspace surgery (scaling all $K - 1$ centroid-pair directions) recovers the predicted $\Delta\,\text{logit}_{\text{global}} / \Delta\,\text{logit}_{\text{single}} \approx d_{\text{eff}}$ in the CIFAR-100 linear regime (ratio= $18.74 \approx d_{\text{eff}} = 24.52$; 3-seed PASS). Two pre-registered cross-domain follow-ups on frozen NLP embeddings failed quantitatively (ratio$\approx$ 2–4 $\ll d_{\text{eff}} =$ 33–54); the directional ordering (global>single) was confirmed. The $1/d_{\text{eff}}$ mechanism appears CNN-specific. Full details in Appendix C.
4. **Ceiling effects.** High-accuracy regimes flatten logits and attenuate measurable intervention slopes.
5. **Large-K regime.** Both ViT and CNN at $K = 100$ show $r \approx 0.75$, below the Monte Carlo noise floor of $E[r] = 0.875$. CIFAR-100's hierarchical superclass structure creates non-ideal kappa geometry; a richer model accounting for class density is needed for large-K settings.
6. **Small-K regime ($K \leq 3$).** The $A(K) = 1.164/\log K + 1.301$ formula was fit on $K =$ 4–28; extrapolation to $K = 2$ or 3 gives unreliable quantitative predictions. Post-hoc tests on sst2 ($K = 2$), tweet_sentiment ($K = 3$), and emotion ($K = 6$) confirm positive Pearson r (0.45, 0.52, 0.42) — the functional form holds — but the predicted magnitude deviates by $\sim$ 70%.
7. **Assumption dependence.** The proof relies on concentration and shared-scale Gumbel approximations; tighter non-asymptotic bounds remain open.
8. **External replication.** Results are internal to the present pipeline and should be independently reproduced.

9. **Multilabel regime.** go_emotions ($K = 28$, multilabel) has $q_{\text{raw}} \approx 0.13$–$0.21$ (well below the single-label Gumbel race regime), and systematically fails cross-dataset tests (one-point calibration $\rho = 0.118$; K-scaling rho below threshold). The CTI law in its current form applies to single-label classification settings; multilabel settings require a separate treatment.
10. **Biological neurons.** Law form confirmed in 30/32 Allen sessions and per-image across three systems; substrate-independent constants rejected ($A_{\text{bio}} \approx 15$–$34\times$ smaller than A_{NLP}). H4 tests fail; full substrate-independence test requires LOAO-style analysis treating sessions as architectures.
11. **Beyond-1NN scope.** κ_{nearest} predicts $k = 3, 5$ NN accuracy ($r \approx 0.79$) and silhouette ($r = 0.88$) pooled; pre-registered primary criterion $r(\kappa_{\text{nearest}}, q_{\text{linear}}) > 0.7$ fails within each dataset ($r = 0.22$–0.59). Linear-probe accuracy depends on global separability, distinct from nearest-neighbour directional SNR.
12. **Two-component geometry.** Two pre-registered null-space experiments confirm κ_{nearest} is the dominant causal predictor ($r = 1.00$ OLMo-1B in null-space identifiability; $R^2 = 0.92$–0.98 in 3×3 causal factorial). Adding κ_{eff} gives $\Delta R^2 \leq 0.03$; cross-architecture bivariate transfer is strong ($r = 0.988$). See Appendix D for full protocol.

9 Conclusion

We present the CTI Universal Law, derived from extreme-value theory:

$$\text{logit}(q_{\text{norm}}) = \alpha\kappa_{\text{nearest}} - \beta \log(K - 1) + C_0,$$

with functional form proven as a conditional theorem under Gaussian class conditionals and Gumbel domain-of-attraction assumptions. The law is validated across 12 NLP architectures spanning transformers, SSM hybrids, and pure linear RNNs (LOAO $\text{CV}_\alpha = 0.019/0.023$), confirmed externally on SmolLM2-1.7B ($\hat{\alpha} = 2.864$; pre-registered PR1: PASS), in a strictly blind OOD test ($r = 0.817$, $p = 0.013$; PASS), and in a pre-registered expanded hold-out across 11 unseen models and 8 datasets ($n = 77$; all 6 pre-registered criteria pass, $r = 0.879$, MAE$= 0.077$). Cross-modal tests confirm the same functional form in ViT ($R^2 = 0.964$) and ResNet50 CNN ($r = 0.749$); the law extends to biological visual cortex (Allen Neuropixels: 30/32 sessions H1 PASS, all 32 showing positive r). The slope $A(K) = 1.164/\log(K) + 1.301$ enables practical prediction: MAE$\leq$0.08 on all six prospective datasets; 4 probe measurements per new architecture reduce MAE by 86% (uncalibrated 0.806$\rightarrow$calibrated 0.111, $r = 0.932$). Universality is of the *functional form*: the constant α is architecture-family-specific ($\alpha_{\text{NLP}} \approx 2.87$, $\alpha_{\text{ViT}} \approx 0.63$, $\alpha_{\text{CNN}} \approx 4.4$), with cross-task drift documented and honest-scope limitations enumerated in §8. Open directions include a renormalization theory for how constants transform across tasks and modalities. The near-simplex equicorrelation $\rho \approx 0.45$ (CV$=$ 3.9% NLP decoders, CV$=$ 1.65% mouse V1, CV$=$ 3.4% across encoder+decoder+SSM families; `cti_cross_family_equicorr.json`) is now confirmed architecture-universal—holding for encoders (BERT-base, ELECTRA-small) with the same numerical value despite 3–5$\times$ higher α. This decouples two distinct universality questions: (1) near-simplex geometry is substrate-independent; (2) the cross-family shift in α ($\alpha_{\text{decoder}} \approx 2.87$ vs $\alpha_{\text{encoder}} \approx 7$) is not explained by ρ. Identical mean pooling is used throughout, ruling out pooling protocol. ELECTRA-small (discriminative RTD pre-training, bidirectional attention) matches BERT-base (MLM, bidirectional) in α_{emp} (3.84 vs 4.37), ruling out pre-training objective; bidirectionality of the attention pattern is the leading candidate for the renormalization mechanism. Other open directions include a multilabel extension and independent replication.

Reproducibility Statement

All numerical claims are taken directly from repository artifacts: `results/cti_smollm2_loao_corrected.json`, `results/cti_smollm2_ood_prediction.json`, `results/cti_expanded_blind_ood.json`, `results/cti_oneshot_calibration_ood.json`,

results/cti_kappa_nearest_universal.json, results/rwkv_preregistration.json, results/cti_kappa_loao_per_dataset.json, results/cti_vit_cross_modality.json, results/cti_resnet50_cifar100.json, results/cti_vit_cifar100.json, results/cti_noisefloor_analysis.json, results/cti_do_intervention_text.json, results/cti_orthogonal_factorial.json, results/cti_checkpoint_sweep.json, results/cti_k_scaling_prediction.json, results/cti_one_point_calibration.json, results/cti_crossdataset_topk.json, results/cti_prospective_new_datasets.json, results/cti_prospective_k_sweep.json, results/cti_k_boundary_sweep.json, results/cti_global_vs_single_surgery.json, results/cti_beyond_1nn.json, results/cti_crossdomain_surgery.json, results/cti_nlp_linear_regime_surgery.json, results/cti_kappa_eff_identifiability.json, results/cti_2d_causal_surface.json, results/cti_extended_family_loao.json, results/cti_neuroscience_cadieu2014.json, results/cti_stringer2018b_validation.json, results/cti_allen_neuropixels_validation.json, results/cti_allen_all_sessions_complete.json, results/cti_spread_vs_K.json, results/cti_spread_vs_K_n10.json, results/cti_kappa_universal_comprehensive.json, results/cti_beta_constrained.json, results/cti_multi_k_sweep.json, results/cti_k10_kappa_sweep.json, results/cti_neff_pairwise_extended.json, results/cti_all_models_fit.json, results/dose_response_preregistration.json, results/cti_equicorrelation_deff.json, results/cti_equicorr_K_sweep.json, results/cti_allen_equicorrelation.json, results/cti_competitive_geometry_verification.json, results/cti_cross_family_equicorr.json, results/cti_competitor_weight_map.json, results/cti_alpha_kappa_rank_corr.json, results/cti_weight_map_transfer.json, results/cti_weight_map_transfer_prereg.json, results/cti_two_knob_identifiability_keff.json, results/cti_preregistered_causal_intervention.json, results/cti_cifar_antitriplet.json, results/cti_confusion_gumbel_test.json, results/cti_confusion_causal_prediction.json, results/cti_cross_arch_abcs.json, results/cti_phi_upgrade_pooled.json, results/cti_kappa_eff_held_out.json, results/cti_kappa_tournament.json, results/cti_j1j2_factorial_rct.json, results/cti_phi_jacobian_synthetic.json, results/cti_phi_jacobian_real.json, results/cti_abcs.json, results/cti_k_scaling_abcs.json, results/cti_bge_dbpedia14_abcs.json, results/cti_null_control_abcs.json, results/cti_utility_revised.json. Preregistration timestamps are verifiable via git log. Code is available at https://github.com/dl1683/ai-moonshots.

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A Derivation Details

Starting from the Gumbel race identity (Eq. 4), under nearest-class dominance:

$$q \approx \left[1 + (K-1)\exp\left(-\frac{\Delta_k}{s}\right)\right]^{-1}.$$

Taking logits:

$$\mathrm{logit}(q) = \frac{\Delta_k}{s} - \log(K-1).$$

With $\Delta_k = a_0 + a_1\kappa_{\text{nearest}} + o(1)$:

$$\mathrm{logit}(q) = \underbrace{\frac{a_1}{s}}_{\alpha}\kappa_{\text{nearest}} - \underbrace{1}_{\beta_\infty}\log(K-1) + \underbrace{\frac{a_0}{s}}_{C_0} + o(1).$$

Finite-data corrections renormalize $\beta_\infty = 1$ to $\beta \approx 0.746$, yielding Eq. 1.

B Pre-registration Deviation Log

Table 4: Pre-registration deviation log. Commits with timestamps are verifiable via `git log --oneline`.

Commit	Change	Justification
`8250bf1`	Dataset substitution: TREC (K=6) → dair-ai/emotion (K=6); PolyAI/banking77 → mteb/banking77	The original datasets require legacy loading scripts deprecated by the `datasets` library (`trust_remote_code=True` no longer accepted). Replacement datasets have identical K values; the pre-registered $A(K)$ predictions (1.951 and 1.569) are unchanged.
`6497907`	Stratified sampling in amazon_massive K=60 test; rare-class guard in `compute_q`	Random subsampling of 1500 from 1265 available stratified samples left some classes with < 2 members, causing `StratifiedShuffleSplit` to fail. The fix (draw $\geq$20 samples per class; drop classes with < 2 members before CV) is a numeric stability patch with no effect on the analysis model or the frozen $A(K)$ prediction.
`7f4c9f8`	RWKV acceptance interval $[2.43, 3.29]$: pre-registration JSON describes this as "mean $\pm$ 3*std(0.054)", but $3 \times 0.054 = 0.162$ gives $[2.727, 3.051]$, not $[2.43, 3.29]$.	The interval half-width 0.43 actually equals $\approx 8\sigma$ of the locked std, not 3σ. This is a documentation error in the pre-registration artifact; the locked interval $[2.43, 3.29]$ and the observed RWKV $\hat{\alpha} = 2.887$ are both as stated. The outcome is unaffected: 2.887 lies within any reasonable interval around 2.889 (the 11-arch prior mean). The paper text now correctly states "wider than $\pm 3\sigma$" (§5.2).

Neither edit changes the pre-registered predictions (A values, hypothesis thresholds, or evaluation metrics); both are verifiable as non-analysis-changing by inspecting the git diffs.

C Surgery Details

Initial attempts to increase κ_{nearest} via CE + triplet-loss fine-tuning with naïve $\lambda = 0.5$ caused representation collapse (within-class variance exploded, q dropped $0.70 \rightarrow 0.22$; the logged collapse metric was the Fisher scatter ratio S_B/S_W, which differs from κ_{nearest}). With L2-normalisation and $\lambda = 0.1$, a pre-registered causal intervention (commit `94ac8a3`; `cti_preregistered_causal_intervention.json`) produced a clean bidirectional training manipulation: CE arm $\text{mean}_q = 0.654$ vs. triplet arm $\text{mean}_q = 0.868$ ($\Delta q = +0.213$, 5/5 seeds direction-correct, PARTIAL PASS); an anti-triplet arm (commit `2b7a812`; `cti_cifar_antitriplet.json`) reduces q to 0.12–0.22 across all 5 seeds (below CE baseline). This bidirectional training manipulation independently confirms that κ_{nearest} is causally active in both directions. The quantitative CTI prediction underestimates by $\approx 5\times$ (predicted $\Delta\,\text{logit} = 0.225$, actual$= 1.244$): training manipulations alter the full representation geometry (not only κ_{nearest}), so the marginal $\alpha \cdot \Delta\kappa$ formula does not account for all confounds in a training-based setting. Single-direction frozen surgery (scaling ONE centroid-pair direction) underperforms the full CTI prediction by $\approx 16\times$: this is understood as a predicted $1/d_{\text{eff}}$ attenuation (a single dimension out of $d_{\text{eff}} \approx 24$). A pre-registered global surgery test (scaling ALL $K-1$ signal-subspace directions; commit `1abdef6`) recovers the full predicted effect in the *linear regime* (CIFAR-100, $\kappa_{\text{eff}} \approx 1.0$, 3

seeds): ratio $\Delta\,\mathrm{logit}_{\mathrm{global}} / \Delta\,\mathrm{logit}_{\mathrm{single}} = 18.74 \approx d_{\mathrm{eff}} = 24.52$ (within-15% at $r = 1.5$). Two pre-registered cross-domain follow-ups on frozen NLP embeddings failed H1: (1) sub-linear regime ($\kappa_{\mathrm{eff}} = 0.24$–$0.73$): median ratio$\approx 2 \ll d_{\mathrm{eff}} \approx 33$; (2) pythia-1b, $\kappa_{\mathrm{eff}} = 1.33$: median ratio$\approx 3.68 \ll d_{\mathrm{eff}} = 54$. Global surgery consistently outperforms single-direction surgery in NLP (directional H2 passes: 10/13 and 5/5), but the quantitative d_{eff} prediction fails. The $1/d_{\mathrm{eff}}$ attenuation mechanism appears specific to the trained CIFAR-100 CNN context.

D Two-Component Geometry Details

Null-space identifiability (commit `6938940`; `cti_kappa_eff_identifiability.json`): null-space scaling provably holds κ_{eff} invariant while changing $\kappa_{\mathrm{nearest}}$ ($< 0.001\%$ variation, H1: 3/3); q tracks $\kappa_{\mathrm{nearest}}$ (OLMo-1B: $r = 1.00$), not κ_{eff} (H4: 1/3 PASS).

2D causal surface (commit `2bc7da6`; `cti_2d_causal_surface.json`): a 3×3 factorial orthogonally manipulates both components on 2 NLP architectures. A bivariate model $\mathrm{logit}(q) = \alpha_1 \kappa_{\mathrm{nearest}} + \alpha_2 \kappa_{\mathrm{eff}} + C$ does not significantly outperform the univariate $\kappa_{\mathrm{nearest}}$ law (H4 FAIL: $\Delta R^2 = 0.03/0.001$; $R^2 = 0.92$–0.98 for $\kappa_{\mathrm{nearest}}$ alone). Cross-architecture bivariate transfer is strong (H6: $r = 0.988, 0.967$). A dimensional confound limits the factorial: in $d = 2048$ with $\approx$2029 null dimensions, compensated signal-scaling redistributes high-dimensional null noise. Combined, these experiments establish $\kappa_{\mathrm{nearest}}$ as the dominant causal predictor; d_{eff} adds marginal information.

Two-knob identifiability (commit `7def904`; `cti_two_knob_identifiability_keff.json`): the most direct test of whether $\kappa_{\mathrm{eff}} = \kappa_{\mathrm{nearest}}\sqrt{d_{\mathrm{eff}}}$ is the sufficient statistic. On CIFAR-100 ($K = 20$, linear-regime checkpoint, 3 seeds), two independent surgical knobs are applied: (A) centroid scaling to change $\kappa_{\mathrm{nearest}}$ while holding d_{eff} fixed; (B) within-class variance redistribution to change d_{eff} while holding $\kappa_{\mathrm{nearest}}$ fixed. If κ_{eff} were sufficient, matched pairs (same κ_{eff}, different κ/d_{eff} decomposition) should give equal $\mathrm{logit}(q)$. **Result (6/7 criteria fail):** mean $|\Delta\,\mathrm{logit}| = 0.30$ across 33 matched pairs (threshold ≤ 0.08); only 24% of pairs meet the invariance criterion (threshold $\geq 80\%$). Crucially, $\mathrm{corr}(\mathrm{residual}, d_{\mathrm{eff}}) = -0.790$: the d_{eff} surgery systematically overpredicts q, while the $\kappa_{\mathrm{nearest}}$ surgery correctly tracks q. **Interpretation:** $\kappa_{\mathrm{nearest}}$ alone is the causal driver; the $\sqrt{d_{\mathrm{eff}}}$ factor in the LOAO formula captures cross-architecture variation observationally but is not independently causally active when manipulated in isolation.

E Additional Tables

Table 5: Global fit statistics (single C_0 and per-dataset C_0).

Quantity	Single C_0	Per-dataset C_0
$\hat{\alpha}$	2.8660	1.4773
$\hat{\beta}$	0.7459	0.3262
R^2	0.6273	0.9552
LOAO CV$_\alpha$	0.0191	0.0228
LODO CV$_\alpha$	0.4193	–

Table 6: RWKV pre-registration and outcome.

Item	Value
Locked prior mean (11 arch)	2.8886
Locked prior std	0.0540
Pre-registered acceptance interval	$[2.43, 3.29]$
Observed RWKV LOAO α	2.8869
Outcome	**PASS**

F Competition Field: Extended Results

F.1 Competition field: j2 is causally active and effects are additive

The original CTI law uses only κ_{j1} (nearest competitor). A 6-arm pre-registered RCT on frozen DBpedia-14 embeddings (pythia-160m, $K = 14$) revealed that the *second-nearest* competitor j_2 is also causally active: single-competitor shift with j_1 fixed gives $r(\kappa_{j2}, \text{logit}\, q) = 0.811$, exceeding the $r < 0.20$ sufficiency threshold. The full competition field $r(\kappa_{\text{all}}, \text{logit}\, q) = 0.988$.

Upgraded law via soft-min competition field. We fit $\phi(\tau) = -\tau \log \sum_{j \neq i} \exp(-\kappa_{ij}/\tau)$ (log-sum-exp soft-min of all competitors) as a replacement predictor for κ_{j1}. A pooled tau-sweep across 5 architectures (70 data points, `cti_phi_upgrade_pooled.json`) finds $\tau^* = 0.20$ (best R^2), yielding $R^2 = 0.461$ vs. $R^2 = 0.360$ for κ_{j1} alone (+28.1% relative improvement), with an *out-of-sample* test on 4 held-out architectures confirming the transfer: pooled R^2 improves from 0.451 to 0.578 (+28.3%, 4/4 architectures pass, `cti_kappa_eff_held_out.json`).

Limitation (functional form specificity): A pre-registered tournament test compared causal weights $\{w_r\}$ against 1000 random monotone weight assignments. The causal weighting achieves $R^2 = 0.578$ and exceeds 67.6% of random nulls, but does not exceed the 90th-percentile random baseline ($R^2_{\text{p90}} = 0.585$; pre-registered threshold; FAIL, `cti_kappa_tournament.json`). The *direction* of the competition gradient (near competitors matter more) is validated; the *specific* exponential functional form $\exp(-\text{gap}/\tau^*)$ is not yet decisively distinguished from arbitrary monotone weighting on this dataset.

Additivity RCT. A pre-registered factorial RCT (`cti_j1j2_factorial_rct.json`) jointly perturbed j_1 and j_2 over DELTA $\in \{0.0, 0.1, 0.2, 0.3, 0.5, 0.8, 1.0\}$. *Additivity*: $r(\Delta\,\text{logit}_q, \Delta\kappa_{j1} + \Delta\kappa_{j2}) = 0.9964$ (n=84 class-delta pairs), confirming effects sum linearly. Mean relative weight $\bar{w} = 0.569$ (Codex prediction: 0.40; threshold: $|w - 0.40| < 0.20$; PASS). Near-tied pairs (gap≈0.004) yield $w \approx 6.4$, while wide-gap pairs give $w \approx 0$, consistent with the phi model: $w \propto \exp(-\text{gap}/\tau^*)$.

Synthetic Jacobian validation. A synthetic Gaussian experiment ($K = 14$, $d = 50$, $N = 2000$; `cti_phi_jacobian_synthetic.json`) fixes $\kappa_{j1} = 0.40$ and varies gap $\in \{0.15, 0.25, 0.35, 0.50, 0.70, 1.00\}$. Results: $r(\log w, -\text{gap}) = 0.9502$ (threshold > 0.85; PASS); fitted $\tau^*_{\text{median}} = 0.207$ (empirical $\tau^* = 0.20$; error 3.5%). This provides a first-principles derivation of τ^* from the geometry.

Real-embedding Jacobian (inconclusive): A matched-delta Jacobian test on pythia-160m step-512 checkpoint embeddings (`cti_phi_jacobian_real.json`) found only 2 of 14 classes meeting both non-ceiling ($q < 0.85$) and rank-stable (gap> 0.04) criteria; the test was inconclusive due to insufficient regime coverage. Qualitatively, wide-gap classes (gap $>$ 0.50) correctly show $w \approx 0$, consistent with phi model predictions; quantitative validation of the exponential functional form in real embeddings remains open.

Confusion-matrix Gumbel test. We test the Gumbel race mechanism directly via the 1-NN confusion matrix C_{ij} (fraction of class i examples whose nearest neighbor is from class j, estimated by 5-fold cross-validation). The Gumbel race predicts $\log C_{ij} \approx \text{const}_i - \kappa_{ij}/\tau_{\text{err}}$, i.e. confusion decays exponentially with kappa (`cti_confusion_gumbel_test.json`). **Pre-registered H1 PASS**: pooled Pearson $r(\log C_{ij}, -\kappa_{ij}) = 0.750$ ($n = 151$ non-zero pairs, threshold 0.50). **Pre-registered H4 PASS**: pooled Spearman $\rho(-\kappa_{ij}, C_{ij}) = 0.768$ (threshold 0.50). The test reveals a *two-level tau structure*: (1) within each class, confusion drops sharply (per-class median $\tau_{\text{err}} \approx 0.026$, winner-take-all: only the nearest competitor causes appreciable confusion); (2) across classes, the pooled best-fit temperature is $\tau_{\text{pooled}} = 0.201 \approx \tau^*$, consistent with the phi model's $\tau^* = 0.20$. This resolves an apparent tension: $\tau^* = 0.20$ captures between-class scaling (why low-kappa classes are confused more) while

$\tau_{\text{err}} \approx 0.025$ captures the within-class sharpness (why κ_{nearest} alone is the sufficient statistic). H2/H3 (requiring per-class $\tau_{\text{err}} \in [0.10, 0.50]$) fail, confirming the local competition is sharper than the phi model's global fit suggests.

Causal confusion-matrix prediction (pre-registered). We run the decisive causal test proposed by Codex: shift the nearest competitor j_1 by $\delta \in \{1.0, 2.0, 3.0\}$ embedding units, predict the *full* confusion-matrix update using *fixed* $A = 1.054$ and $\tau^* = 0.20$ (no refit), then compare to the measured post-intervention confusion (`cti_confusion_causal_prediction.json`).

Pre-registered H1: pooled $r(\hat{C}_{\text{pred}}, C_{\text{actual}}) > 0.50$ using $\tau^* = 0.20$. Pre-registered H3: sign accuracy of $\Delta C_{i,j1}$ (predicted direction of confusion change) ≥ 0.80.

All criteria pass at all three deltas:

- $\delta = 1.0$: $r_{\tau^*} = 0.842$ (PASS), sign acc.= 92.9% (PASS)
- $\delta = 2.0$: $r_{\tau^*} = 0.808$ (PASS), sign acc.= 100% (PASS)
- $\delta = 3.0$: $r_{\tau^*} = 0.776$ (PASS), sign acc.= 100% (PASS)

Importantly, $\tau^* = 0.20$ outperforms $\tau_{\text{local}} = 0.026$ for causal prediction ($r = 0.842$ vs $r = 0.683$ at $\delta = 1.0$), confirming that τ^* captures causal response sensitivity while τ_{local} describes the equilibrium confusion structure. Predicted per-class q values match observed within 0.01–0.07 logit units (no refit), providing quantitative causal-predictive validation of the CTI framework.

Unconfounded competitor weight map (ci-fixed interventions). The competition experiments above shift c_j while c_i may move under projection; a fully unconfounded estimate requires anchoring c_i at its original position while moving only c_j. We run 14 classes $\times$ 13 competitors $\times$ 11 delta points $\times$ 5-fold CV with c_i fixed (`src/cti_competitor_weight_map.py`; `results/cti_competitor_weight_map.json`). The per-rank causal weight $w_r = \alpha_r / \alpha_{j_1}$ (normalised by nearest-competitor slope) yields the profile $w = [1.00, 0.29, 0.21, 0.17, 0.09, 0.08, 0.03, \ldots]$: a heavy-head distribution where only ranks 1–4 are causally significant. The summed effective competition $\sum_r w_r = 2.42$ is consistent with $\alpha_2 = 1.523$ (2-competitor localisation; §F.1), independently confirming that 2–4 near competitors drive the cross-architecture constant. The nearest-competitor slope $\alpha_{j_1} = 1.272$ is close to (but below) LOAO $\alpha = 1.477$; the gap reflects residual saturation in high-accuracy DBpedia classes.

Alpha-kappa rank correlation: monotone decay confirmed. As an internal validation that competition weight decays with kappa distance (as theory requires), we compute Spearman $\rho(\alpha_r, \text{kappa_rank})$ for each of the 14 classes (`results/cti_alpha_kappa_rank_corr.json`). **14/14 classes show negative correlation** (sign test $p = 6.1 \times 10^{-5}$, Wilcoxon $p = 1.2 \times 10^{-4}$); mean $\rho = -0.551$, 95% CI $[-0.665, -0.436]$. This confirms that competition weight decays monotonically with kappa distance, as predicted by the Gumbel race theory, without relying on any functional-form assumption.

Cross-architecture weight transfer (pre-registered; negative result). We pre-registered a no-refit transfer test: use the pythia-160m weight profile w_r to predict total competition effects $\alpha_{\text{top}}\, m$ on two held-out architectures (pythia-410m, bert-base; `results/cti_weight_map_transfer_prereg.json`, pre-registered commit `c013ddc`). Both H1 (relative error $< 30\%$) and H2 (Pearson $r > 0.90$) fail on both architectures. Root cause: the held-out architectures operate in a saturation regime ($q > 0.85$) on DBpedia K=14, making $\alpha_{j_1} = \delta\,\text{logit}/\delta\kappa \to \infty$ and rendering the normalised weight profile unidentifiable. The pythia-160m source slope $\alpha_{j_1} = 1.272$ vs. pythia-410m $\alpha_{j_1} = 12.6$ and bert-base $\alpha_{j_1} = 9.6$ reflects saturation amplification rather than a genuinely different competition structure. We record this as a pre-registered negative result: w_r cross-architecture universality cannot be evaluated on a dataset where held-out models are near ceiling. The finding motivates future transfer tests on harder datasets ($q \in [0.5, 0.8]$).

Table 7: Competition field evidence summary.

Experiment	Result	Status
j2 causal activity (6-arm RCT)	$r = 0.811$	Causally active
Full field activity	$r = 0.988$	PASS
Phi upgrade pooled R^2	$0.360 \to 0.461$ (+28.1%)	PASS
Held-out arch transfer (4/4)	$0.451 \to 0.578$ (+28.3%)	PASS
Monotone null (causal vs random weights)	67.6% beat, p90 FAIL	FAIL
Additivity RCT	$r = 0.9964$	PASS
Synthetic Jacobian $r(\log w, -\text{gap})$	0.9502	PASS
Synthetic τ^*	0.207 vs. empirical 0.20	3.5% error
Real-embedding Jacobian	2/14 valid classes	INCONCLUSIVE
Confusion matrix Gumbel (r, ρ)	0.750, 0.768	PASS (H1+H4)
Causal confusion prediction ($\delta = 1..3$, no refit)	$r = 0.842$–0.776	PASS (H1+H3)
Unconfounded weight map (ci-fixed)	$\sum_r w_r = 2.42$; w_1–w_4 significant	PASS
α-kappa rank correlation	14/14 negative, $\bar{\rho} = -0.551$, $p = 1.2 \times 10^{-4}$	PASS
Cross-arch weight transfer (pre-reg.)	H1/H2 FAIL (saturation regime)	NEGATIVE
Cross-arch A1 scaling (5 archs, 3 families)	r-scale ≥ 0.90 (5/5; min observed 0.961)	PASS
Null-space surgery control	slope ratio 1.4% of competitive	PASS
K-scaling ABCS ($K = 6, 9, 10, 20$)	$r(K, K_{\text{eff}}) = 0.998$; A1 4/4	PASS
BGE-small encoder ABCS ($K = 14$)	r-scale $= 0.981$; $K_{\text{eff}} = 13$	PASS

G Composite Kappa-K Sweeps: Detailed Results

G.1 Composite Kappa-K Law: Multi-K and K=10 Sweeps

The comprehensive universality test (above) sweeps architecture at fixed K per dataset. A complementary validation sweeps κ at fixed K: how does q change as the nearest-class gap varies, and does the $\sqrt{K-1}$ competition scaling persist at different K?

B(κ) power-law form. Define $B = q_{\text{base}}/(1 - q_{\text{base}})$ as the baseline performance ratio when only the minimum κ class pair is varied. The Gumbel race predicts:

$$B(K, \kappa) = C_0 \frac{\sqrt{K-1}}{\kappa^p}, \quad p > 0, \quad C_0 = \sqrt{2/3} \approx 0.816, \tag{7}$$

where p is the empirical kappa exponent and C_0 is derived analytically from the Gumbel race variance.

Multi-K sweep ($K \in \{5, 7, 10, 14, 20\}$, pre-registered). The pre-registered multi-K sweep (`src/cti_multi_k_sweep.py`) varies K at a fixed target κ and measures B. Result (`results/cti_multi_k_sweep.json`): $B \propto \sqrt{K-1}$ with CV $= 4.4\%$ across 5 K values and 3 alpha levels — the $\sqrt{K-1}$ scaling is confirmed with high consistency. This is independent evidence for $N_{\text{eff}} \propto \sqrt{K-1}$ using synthetic Gaussian-mixture embeddings (not real neural networks).

K=10 kappa sweep (pre-registered). The $K = 10$ sweep (`src/cti_k10_kappa_sweep.py`, 6 kappa values $\kappa \in \{0.15, 0.20, 0.25, 0.30, 0.40, 0.50\}$, 3 alpha levels, 12 seeds) tests whether Eq. 7 holds at $K = 10$ with the specific C_0 derived from the $K = 7$ strong pass.

Results (`results/cti_k10_kappa_sweep.json`): **STRONG PASS** (all 5 pre-registered criteria):

- **PR1** (all $|\text{rel_err}| < 30\%$): 6/6 kappas pass. Maximum relative error $= 19\%$ (kappa$=$ 0.20).
- **PR2** (log-log slope $\in [-0.90, -0.55]$): slope$= -0.739$. PASS.
- **PR3** (C_0 within $\pm 20\%$ of prior 0.816): $C_0 = 0.773$, relative error $= 5.4\%$. PASS.
- **PR4** (C_0 CV $< 25\%$): CV$= 4.2\%$. PASS.
- **PR5** (Pearson $p < 0.05$): $r = -0.992$, $p < 10^{-4}$. PASS.

Table 8: Beta-constrained model comparison: LOAO and LODO MAE across 444 data points, 19 architectures, 10 datasets. Winner (M2, $\beta = 0.5$) is highlighted.

Model	β	LOAO MAE	LODO MAE	Status
M2 (sparse)	**0.5**	**0.766**	**0.857**	**WINNER**
M0 (unconstrained)	0.478	0.771	0.898	+0.6% vs M2
M3 (K-indep.)	0.0	0.917	0.899	
M1 (dense ETF)	1.0	0.852	1.069	+10.1% vs M2

The $C_0 = 0.773$ at $K = 10$ is 5.4% below the prior $C_0 = 0.816$ (K=7 result), consistent with easy-class dilution: at $K = 10$, 8 easy classes with large κ drive q_{base} near 1, diluting the κ-surgery signal. This systematic dilution decreases with K (more hard pairs at large K) and with κ (harder pairs have larger signal). The pattern is theoretically expected and confirms the law's functional form without altering the universal scaling.

H Sparse Competition: Full Analysis

H.1 Sparse Competition: Pre-Registered Beta Exponent Test

The Gumbel race identity (Eq. 4) under nearest-class dominance predicts $\beta_\infty = 1.0$ (all $K - 1$ competitors equally active). The empirical global estimate $\hat{\beta} = 0.478$ (positive sign convention; from the 444-point fit) is substantially below 1.0. We tested two competing mechanistic hypotheses:

- **Dense ETF competition (M1)**: $\beta = 1.0$ (all $K - 1$ competitors contribute equally, as in an equiangular tight frame); α and C_0 fit freely.
- **Sparse competition (M2)**: $\beta = 0.5$ (only $N_{\text{eff}} \propto \sqrt{K - 1}$ competitors contribute); α and C_0 fit freely.

We additionally tested M0 (all free) and M3 ($\beta = 0$, K-independent).

Pre-registration. The model comparison was pre-registered (`src/cti_beta_constrained_test.py`, committed before running; `results/cti_beta_constrained.json`) with criteria: (PR_WIN) M2 LOAO MAE $\leq$ M1 LOAO MAE; (PR_CLOSE) relative improvement $\geq 3\%$; (PR_UNCON_GAP) unconstrained M0 within 5% of winner.

All three pre-registered criteria pass. PR_WIN: M2 MAE $= 0.766 <$ M1 MAE $= 0.852$ (10.1% improvement). PR_CLOSE: PASS ($\geq 3\%$ threshold). PR_UNCON_GAP: M0 (unconstrained, $\hat{\beta} = 0.478$) is within 0.6% of M2, confirming $\beta = 0.5$ is not under-constrained—it is effectively the best global estimate. LODO confirms: M2 beats M1 by 24.8% (0.857 vs. 1.069).

Per-K winner breakdown. M2 ($\beta = 0.5$) wins at $K \in \{4, 6, 10, 59, 77\}$ (sparse, task-specific classes); M1 ($\beta = 1.0$) wins at $K \in \{14, 20, 28, 42\}$ (semantically dense topic/emotion datasets). This is consistent with a regime-dependent competition structure: at low K (few distinct classes) and very high K (many fine-grained task classes), competition is sparse ($\sqrt{K}$ scale); at medium K with semantically overlapping topics (news, emotions), competition approaches the dense ETF limit.

Sparse competition mechanism. The Gumbel race sum (Eq. 4) can be decomposed as $\sum_{j \neq k} \exp(-\Delta_{k,j}/s) \approx N_{\text{eff}} \exp(-\Delta_k/s)$ where N_{eff} is the effective number of competitive classes. The empirical $\beta = 0.5$ implies $N_{\text{eff}} \propto \sqrt{K - 1}$. A plausible mechanism (Codex analysis): in real text classification, classes are organized hierarchically. Competitors at the same semantic level number $\sim \sqrt{K}$; deeper-level competitors are substantially farther away and contribute negligibly. Equivalently, if each competitor activates an effective gate with probability $\propto 1/\sqrt{K}$ (sparse gating), then $N_{\text{eff}} = (K - 1) \cdot p_K \approx \sqrt{K - 1}$.

Triangulated evidence for $N_{\text{eff}} \propto \sqrt{K-1}$. Three independent experimental frameworks converge on the same exponent:

1. **Synthetic sweep (controlled geometry)**: the multi-K sweep (Eq. 7, 5 values $K \in \{5, 7, 10, 14, 20\}$) finds $B(K, \kappa) \propto \sqrt{K-1}/\kappa^p$ with CV= 4.4%. The numerator $\sqrt{K-1}$ arises directly from the Gumbel race competition count.
2. **Real network global fit**: across 444 points (19 architectures, 10 datasets, $K = 4$–77), the unconstrained OLS estimate is $\hat{\beta} = 0.478 \approx 0.5$.
3. **Pre-registered constrained comparison**: $\beta = 0.5$ beats $\beta = 1.0$ by 10.1% LOAO and 24.8% LODO; the unconstrained global estimate lies within 0.6% of $\beta = 0.5$, confirming it is the MLE under the Gumbel race model.

Synthetic and real-network results are mutually independent: (1) uses Gaussian-mixture embeddings with controlled κ/d_{eff}, while (2)–(3) use transformer embeddings on natural-language tasks.

Table 9: N_{eff} geometric exponent and competition density $\rho = N_{\text{eff}}(K{=}14)/(K-1)$ across 7 architectures on DBpedia. Denser inter-class competition (higher ρ) predicts steeper N_{eff} scaling (higher β_{geom}). RWKV (pure linear RNN) validates the mechanism beyond attention-based models.

Architecture	Type	β_{geom}	$\rho = N_{\text{eff}}/(K-1)$
Pythia-160m	Decoder (GPT-NeoX)	0.934	0.804
TinyLlama-1.1B	Decoder (LLaMA)	0.923	0.779
RWKV-169M	Linear RNN	0.861	0.659
OLMo-1B	Decoder (OLMo)	0.806	0.578
Qwen2.5-0.5B	Decoder (Qwen)	0.829	0.564
Qwen3-0.6B	Decoder (Qwen3)	0.791	0.566
GPT-Neo-125M	Decoder (GPT-Neo)	0.711	0.454
Pearson $r(\rho, \beta_{\text{geom}}) = 0.978$ ($p = 0.0001$, $n = 7$); Spearman $\rho = 0.893$ ($p = 0.007$)			

Direct centroid-geometry and beta reconciliation. A direct test (`src/cti_neff_pairwise.py`) extracts all class centroids from Pythia-160m on DBpedia ($K = 4$–14 subsets, $n = 20$ seeds each) and computes $N_{\text{eff,k}} = \sum_{j \neq k} \exp(-\alpha_{\text{univ}}(\kappa_{kj} - \kappa_{k,\text{near}}))$ where $\alpha_{\text{univ}} = 3.598$. Results: DBpedia geometric exponent $= 0.934 \pm 0.005$ ($r = 0.9999$); Banking77 ($K = 4$–64) exponent $= 0.918 \pm 0.012$ ($r = 0.9999$). Across seven architectures (Pythia-160m, TinyLlama-1.1B, RWKV-169M, OLMo-1B, Qwen2.5-0.5B, Qwen3-0.6B, GPT-Neo-125M), the geometric N_{eff} exponent ranges from 0.71 to 0.93 (mean 0.84; see Table 9 and `results/cti_neff_pairwise_extended.json`). The exponent correlates with competition density $\rho = N_{\text{eff}}(K_{\text{full}})/(K_{\text{full}} - 1)$ at the full class count: Pearson $r = 0.978$ ($p = 0.0001$, $n = 7$), Spearman $\rho = 0.893$ ($p = 0.007$). Architectures that create denser, more uniform inter-class separation (Pythia-160m: $\rho = 0.80$, TinyLlama-1.1B: $\rho = 0.78$) exhibit steeper geometric scaling; those with sparser competition (GPT-Neo-125M: $\rho = 0.45$) scale more sub-linearly. Crucially, RWKV (pure linear RNN with no attention) follows the same trend ($\rho = 0.66$, $\beta_{\text{geom}} = 0.86$), confirming the mechanism is architecture-independent.

These geometric exponents (0.71–0.93) exceed the regression-based $\beta = 0.478$; this does *not* represent a contradiction. A direct within-architecture test (`src/cti_beta_reconciliation.py`) measures the univariate slope of $\text{logit}(q_{\text{norm}})$ vs $\log(K-1)$ as K is varied by subsampling classes from a fixed architecture (Pythia-160m). Without controlling for κ_{nearest}: slope ≈ -0.52 (DBpedia), -0.41 (Banking77) — both consistent with $\beta \approx 0.5$. After controlling for κ_{nearest}: slope $\approx +0.03$ (DBpedia), $+0.31$ (Banking77) — κ_{nearest} mediates nearly all K-dependence within a single architecture, especially at small K ranges. The cross-dataset regression $\beta = 0.478$ is an average partial slope across datasets with varied C_0; the geometric N_{eff} exponent measures a distinct quantity (competition density scaling) that is inherently above β because it does not subtract the κ_{nearest} mediation term. Both quantities confirm the sparse-competition regime

($\beta < 1$): the Gumbel race with $N_{\text{eff}} \propto K^{0.81}$, combined with κ_{nearest} mediating K-dependence, produces the observed $\beta \approx 0.5$ in the law.

H.2 H8+ expanded holdout detail

Table 10: H8+ per-model holdout summary (11 unseen architectures). All parameters frozen from 20-model training set (pre-registered commit `719da25`). n is the number of valid dataset predictions per model after floor/ceiling exclusion.

Model	Family	n	MAE	Max err	Status
distilbert-base-uncased	encoder	8	0.036	0.060	Best
albert-base-v2	encoder	7	0.067	0.122	Good
roberta-base	encoder	8	0.071	0.309	Good
falcon-rw-1b	decoder	8	0.051	0.110	Good
stablelm-3b-4e1t	decoder	8	0.070	0.155	Good
opt-125m	decoder	7	0.071	0.148	Good
qwen2.5-1.5b	decoder	7	0.077	0.194	Good
phi-1.5	decoder	8	0.082	0.183	Good
pythia-2.8b	decoder	8	0.082	0.172	Good
bloom-560m	decoder	4	0.160	0.316	Weak
gemma-3-1b	decoder	4	0.167	0.533	Weak
All (full H8+)		77	0.077	0.533	PASS
Excl. bloom+gemma		69	0.067	0.309	—
$q \in [0.05, 0.95]$		73	0.073	0.533	—

bloom-560m operates near chance on 4/8 datasets (all excluded by floor filter or yielding large errors); gemma-3-1b catastrophically mispredicts DBpedia (err $= 0.533$, predicted $q = 0.681$, actual $q = 0.148$). These two models account for all outliers > 0.20. Excluding them yields $r = 0.938$, MAE$= 0.067$ ($n = 69$). The three newly added models (falcon-rw-1b, phi-1.5, qwen2.5-1.5b) all perform well (MAE ≤ 0.082), confirming that the law generalizes beyond the initial 8-model set. The pre-registered test reports the full set; the sensitivity rows are exploratory.